

JINGQUAN YAN

jingquan.yan@uta.com ◇ Arlington, TX

EDUCATION

Ph.D in Computer Science and Engineering, University of Texas at Arlington Expected 2028

GPA: 4.0/4.0

Courses: Data Analysis and Modeling, Distributed System, Parallel Processing, Computer Vision, Advanced Graphics and Image Processing

Computer Science, Rutgers University 2021 - 2023

GPA: 4.0/4.0

Courses: Massive Data Mining, Bayesian Deep Learning, Machine Learning, Linear Programming, Computers in Biomedicine, Learning in Humans and Machines, Foundation of Computer Science

M.S. in Electrical Engineering, University of Southern California 2019 - 2021

GPA: 3.94/4.0

Courses: Linear Algebra, Mathematical Pattern Recognition, Advanced Computer Vision, Optimization, Analysis of Algorithms, Simulation Methods for Stochastic System, Probability for Engineers

B.E. in Communication Engineering, University of Science and Technology Beijing 2015 - 2019

GPA: 3.66/4.0

Dissertation: “Complex Aviation Mobile Services Link Aggregation with Hidden Markov Model”

Supervised by Dr.Bing Du with University Excellent Student Paper Award (top 3%)

PUBLICATIONS

- [1] **Self-Interpretable Time Series Prediction with Counterfactual Explanations** ([link](#))
Jingquan Yan, Hao Wang
ICML 2023 Oral (top 8% accepted)
Keywords: Bayesian Deep Learning, Causal Inference, Time Series, Healthcare, Explainable AI
- [2] **Breaking the Correlation Plateau: On the Optimization and Capacity Limits of Attention-Based Regressors** ([link](#))
Jingquan Yan, Yuwei Miao, Peiran Yu, Junzhou Huang
ICLR 2026 Poster (28% acceptance rate)
Keywords: Joint Optimization, Correlation Learning, Attention Mechanism, Regression Model
- [3] **B2P-Corr: Batch-to-Population Gradient Estimators for Non-Decomposable Correlation Losses** ([link](#))
Jingquan Yan, Yuwei Miao, Peiran Yu, Junzhou Huang
NeurIPS 2026 Poster (25.7% acceptance rate)
Keywords: Mini-batch Optimization, Non-decomposable Objectives, Correlation Learning
- [4] **GenePheno: Interpretable Gene Knockout-Induced Phenotype Abnormality Prediction from Gene Sequences** ([link](#))
Jingquan Yan, Yuwei Miao, Lei Yu, Yuzhi Guo, Xue Xiao, Lin Xu, Junzhou Huang
AAAI 2026 Oral (top 25% accepted)
Keywords: Phenotype Prediction, Gene Ontology, Multi-label Classification, Explainable AI, Label Correlation
- [5] **CytoWave: Perturbation-Centric Pretraining for Single-Cell Response Prediction** ([link](#))
Yuwei Miao, **Jingquan Yan**, Hehuan Ma, Thao M. Dang, Lin Xu, Siyuan Zhang, Junzhou Huang
NeurIPS 2026 Poster (25.7% acceptance rate)
Keywords: Single-cell Pretrained Model, Perturbation Prediction
- [6] **Decomposed Representations Mitigate the Alignment-Specificity Trade-off in Multi-Omics** ([link](#))
Thao M. Dang, Feng Jiang, Hehuan Ma, Yuzhi Guo, **Jingquan Yan**, Haiqing Li, Saiyang Na, Zheng Zheng, Thuc Anh Tran, Jean Gao, Junzhou Huang

NeurIPS 2026 Poster (25.7% acceptance rate)

Keywords: Multi-modality Learning, Learning with Missing Modality

- [7] **GoBERT: Gene Ontology Graph Informed BERT for Universal Gene Function Prediction** ([link](#))

Yuwei Miao, Yuzhi Guo, Hehuan Ma, **Jingquan Yan**, Feng Jiang, Rui Liao, Junzhou Huang

AAAI 2025 Poster (23.4% acceptance rate)

Keywords: Gene Ontology, Gene Function Prediction, Language Model

- [8] **Uncertainty-Aware Multimodal Gait Representation Learning for Scoliosis Screening** ([link](#))

Haiqing Li, **Jingquan Yan**, Yinhao Wu, Yuzhi Guo, Hehuan Ma, Wenliang Zhong, Jean Gao, Junzhou Huang

IEEE Transactions on Medical Imaging 2026-09

Keywords: Gait Recognition, Evidential Deep Learning

- [9] **GAMI: Gradient-Aligned Modality Importance for Multimodal Cancer Outcome Prediction** ([link](#))

Xiao Hu, Yuzhi Guo, Zheng Zheng, Wenliang Zhong, Haiqing Li, Yuwei Miao, **Jingquan Yan**, Hehuan Ma, Yinhao Wu, Junzhou Huang

BIBM 2026

Keywords: Multi-modality Learning, Survival Prediction

- [10] **UniEntrezDB: Large-scale Gene Ontology Annotation Dataset and Evaluation Benchmarks with Unified Entrez Gene Identifiers** ([link](#))

Yuwei Miao, Yuzhi Guo, Hehuan Ma, **Jingquan Yan**, Feng Jiang, Weizhi An, Jean Gao, Junzhou Huang

Preprint: arXiv:2412.12688

- [11] **HiddenVis: a Hidden State Visualization Toolkit to Visualize and Interpret Deep Learning Models for Time Series Data**

Jingquan Yan, Ruichen Rong, Guanghua Xiao, Xiaowei Zhan

Preprint: bioRxiv 2020.12.11.422030

RESEARCH INTERESTS

- **Trustworthy and Interpretable AI:** emphasis on explainable and uncertainty quantification in AI systems
- **Correlation Learning:** learning and exploiting label and feature correlations in modern machine learning
- **AI4Science:** applications of deep learning in healthcare (time-series, pathology data) and biology (genetic data)

SERVICES

Gold Reviewer Award

ICML 2026

Reviewer

ICML, ICLR, NeruIPS, CVPR, AAAI, WACV, TMLR

EXPERIENCE

Full-time Teaching Assistant

University of Texas at Arlington

Fall 2024

Arlington, TX

- CSE3310: Software Engineering (Fall 2024)
- CSE4322: Software Project Management (Spring 2025)

Full-time Teaching Assistant

Rutgers University

Sep 2021 - May 2023

Piscataway, NJ

- CS336: Database Management System (Fall 2021, Spring 2022, Fall 2022)
- CS210: Data Management for Data Science (Spring 2023)

Research Assistant

UT Southwestern Medical Center

May 2020 - Sep 2020

Dallas, TX

- Proposed a semi-supervised deep learning framework for long-term mortality prediction for NHANES accelerometer dataset
- Implemented a neural network hidden state visualization and exploration tool with Flask and JavaScript

Research Assistant

ZTE Corporation

June 2018 - July 2018

Beijing, China

- Researched on robust and secure allocation protocols for MISO NOMA systems
- Implemented an efficient suboptimal “polybox” algorithm to solve complex optimization problem using MATLAB

National Undergraduate Training Programs

University of Science and Technology Beijing

Dec 2016 - Dec 2017

Beijing, China

- Researched on effective mass spectrum and dissipation power of granular material under excitations
- Implemented parallel computing program using LIGGGHTS and ParaView, visualized with Python GUI package

HONORS

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|---|-----------|
| • Silver Medal in Kaggle ASHRAE - Great Energy Predictor III Competition | Dec 2019 |
| • Outstanding Graduate Student in Computer Science and Communication Department | June 2019 |
| • University Excellent Undergraduate Paper Award (top 3%) | June 2019 |

SKILLS

Programming Language

- Proficiency in Python, MATLAB
- Familiar with C++, R, Javascript and LabView

Framework

- Pytorch, Pyro, Huggingface, AutoGen, JAX

Language Proficiency

- Chinese (native), English (fluent)